

11/8/26

CS340

Lecture 6

Size of Subset DFA

In the ~~above~~ ^{subset} construction number of states in D is exponential in the number of states in M . This exponential ~~to~~ blow-up is unavoidable, ~~it~~ follows from the next theorem.

Theorem 1 ^(of lec 5) Let L_k be the language of Example 2. Any DFA to recognize L_k has at least 2^k states.

Proof: Let D be a DFA $(Q_D, \Sigma, q_{0,D}, \delta_D, F_D)$ s.t. $L(D) = L_k$. (Here $\Sigma = \{0, 1\}$).

Towards contradiction assume that $|Q_D| < 2^k$.

Consider $\delta_D(q_{0,D}, x)$ for $x \in \{0, 1\}^k$. By pigeonhole principle, $\exists x, y \in \{0, 1\}^k [x \neq y \wedge \delta_D(q_{0,D}, x) = \delta_D(q_{0,D}, y)]$.

As $x \neq y$, there is an $1 \leq i \leq k$ s.t. i^{th} bit of x (denoted x_i) is not equal to i^{th} bit of y (denoted y_i). Consider strings $x0^{i-1}, y0^{i-1}$.

$$\delta_D(q_{0,D}, x0^{i-1}) = \delta_D(q_{0,D}, y0^{i-1})$$

$$\therefore \delta_D(q_{0,D}, x) = \delta_D(q_{0,D}, y)$$

But exactly one of $x0^{i-1}, y0^{i-1} \in L_k$.

Therefore, $L(D) \neq L_k$, giving us the desired contradiction $\square$

Question: Can we close the gap between upper bound (on the no. of states in DFA of ~~subset construction~~ ^{subset construction}) and lower bound of the theorem above? That is, either show an NFA for L_k with k states or improve the lower bound in the above theorem to 2^{k+1} .

Ans: Consider the DFA constructed by subset construction for NFA

of $k+1$ states recognizing L_k . Consider the states of this DFA that are reachable from its initial state. How many states are there?

Example: Construct ^{directly} a DFA with 2^k states which recognizes L_k .

Solution $M = (\Sigma^k, \Sigma, \emptyset, \delta, 1 \Sigma^{k-1})$

$$\delta(a_1 \dots a_k, a) = a_2 a_3 \dots a_k a \quad \square$$

In many practical applications DFA constructed by subset construction has much smaller size than $2^{|\mathcal{Q}|}$. That is, set of reachable sets of this DFA from its initial state is much smaller than $2^{|\mathcal{Q}|}$. We discuss one such application below.

Pattern matching

In this problem we are given a text $\in \Sigma^*$ and a pattern $\in \Sigma^*$. The text is usually much longer than the pattern, one needs to find all occurrences of the pattern in the text string.

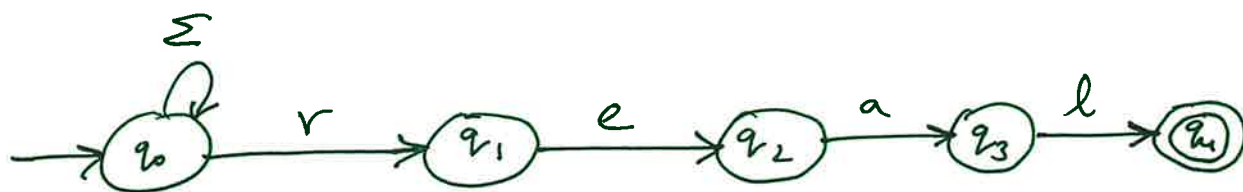
We can use automate theoretic approach to solve this problem.

Consider the following algorithmic problem.

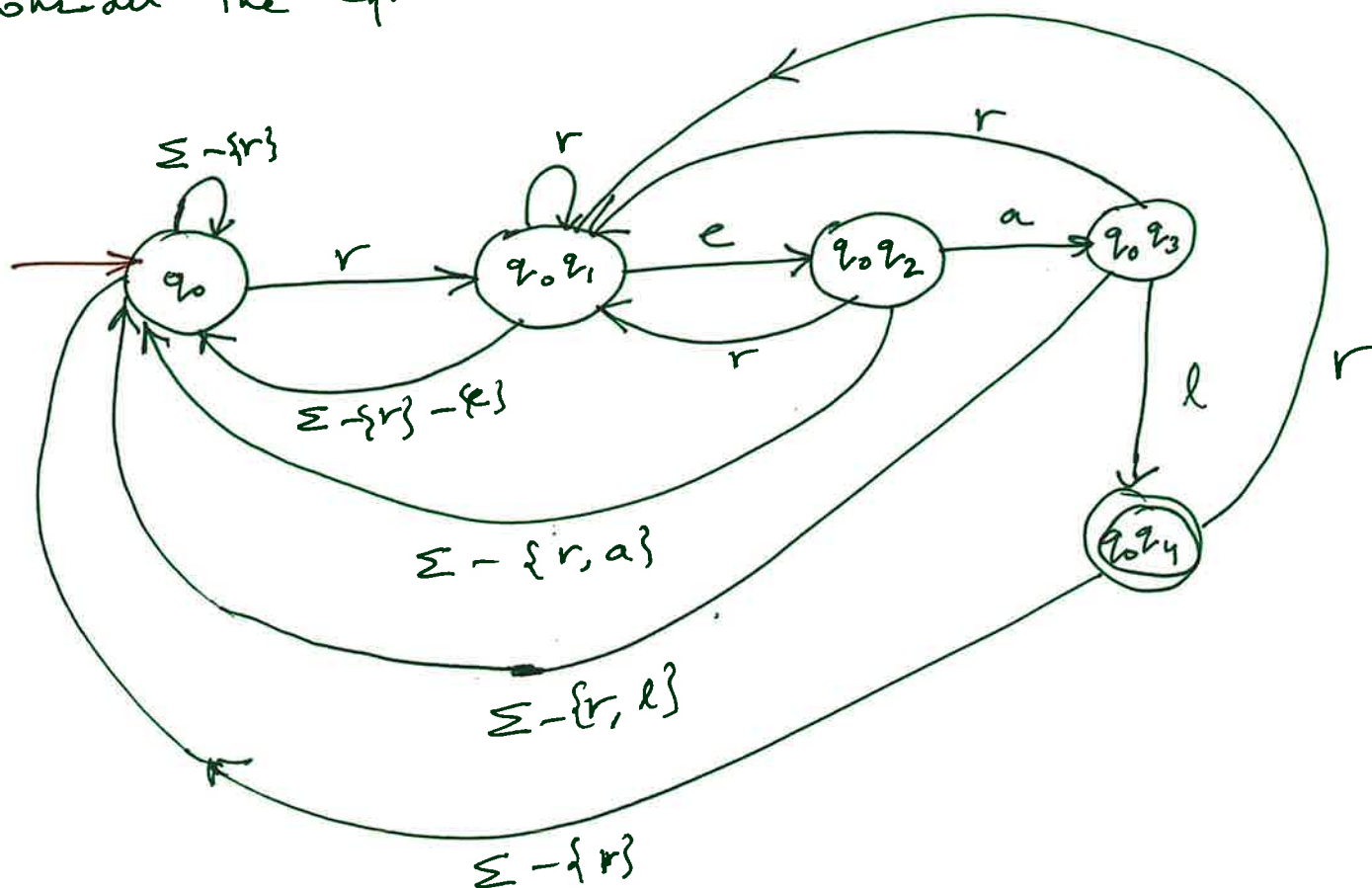
Input: $w \in \{a, b, \dots, y, z\}^*$

output: Find all positions in w where ~~occurrence of~~ word "real" occurs

To solve this we design a NFA which accepts exactly those strings over $\Sigma = \{a, b, \dots, y, z\}$ which end in letters ~~'real'~~ sequence "real".



Consider the equivalent DFA for this NFA.



Note that this DFA constructed using reachable part of subset automata has just 5 states (instead of 2^5 states).

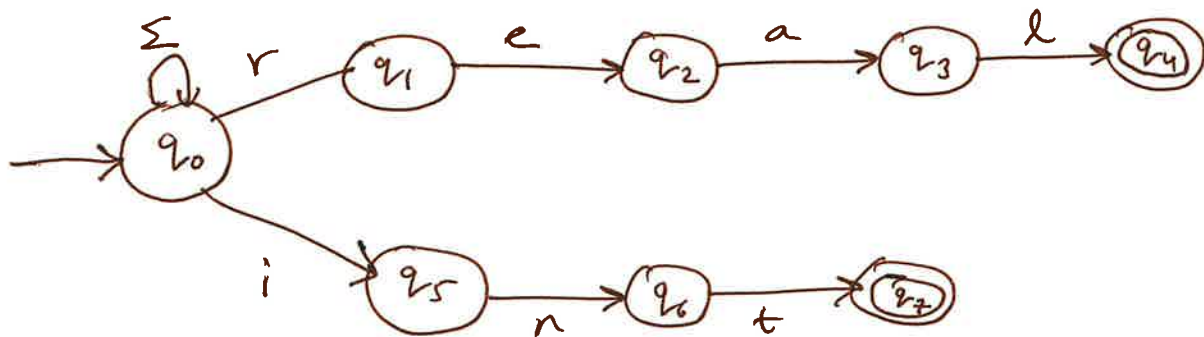
As DFA is equivalent to NFA. Language recognized by the DFA is $\Sigma^*.\{\text{real}\}$.

We can simulate in time $O(|w|)$ the computation of this automata on w . (transition table of the DFA may be stored as a two dimensional array).

Whenever this simulation indicates that DFA is in final state then we know that the last four letters in the text were "real".

Example: Consider the problem of finding all occurrences of patterns "real", "int" in a given text over $\Sigma = \{a, \dots, z\}$.

We design NFA for this problem as follows.



We show (reachable part of) subset automata for this NFA on the next page.

